Soil and Water Conservation Department

“दुष्काळ हा भुतकाळ
जलसमृध्दी हा भविष्यकाळ”

Maharashtra's starting point

89% households in Maharashtra[1] have access to tap water connections

Maharashtra has rapidly expanded its irrigation from 4 Lakh ha. in 1960[2] to 104 Lakh ha. in 2024

Maharashtra ranks third in micro-irrigation, with 23 lakh hectares[3]

The state treats 46% of wastewater to primary levels, above the national average of 28%[4]

1. Jal Jeevan Mission
2. Kelkar Committee Report
3. Directorate of Economics and Statistics, 2023, Karnataka leading with 29 lakh ha followed by Rajasthan with 25 lakh ha under micro irrigation
4. Central Pollution Control Board, 2021; Open Government Data Platform, 2022; Centre for Water and Sanitation, 2024

Key Challenges

Maharashtra has **only 32% Jal Jeevan Mission certified** villages[1]

16% of ground water blocks are semi-critical, critical or over-exploited[2], 30% monitoring stations are classified as 'polluted'

Fragmented institutional structure: WRD, WCD, WSSD, GSDA, and Agriculture departments lack convergence

Irrigation as share of total cropped area is 43% in Maharashtra against average gross irrigated area of 60% in India[3]. Only 70–75% is potential is utilized, as compared to 84% in India

15% of water stored in the surface irrigation projects is lost due to evaporation and 30% of water is lost during conveyance[4]. **Regional Imbalance Across River Basins**

Maharashtra's water treatment levels (46%) are **lower than developed countries** in Western Europe (86%)[5]. Only 4% of treated wastewater is reused

1. Jal Jeevan Mission; 2. Central Ground Water Board, Dynamic Ground Water Resources of India, 2024; 3. Directorate of Economics and Statistics, 2024, Land use statistics at a glance; 4. Water audit report 2022-23; 5. Earth System Science Data, 2020

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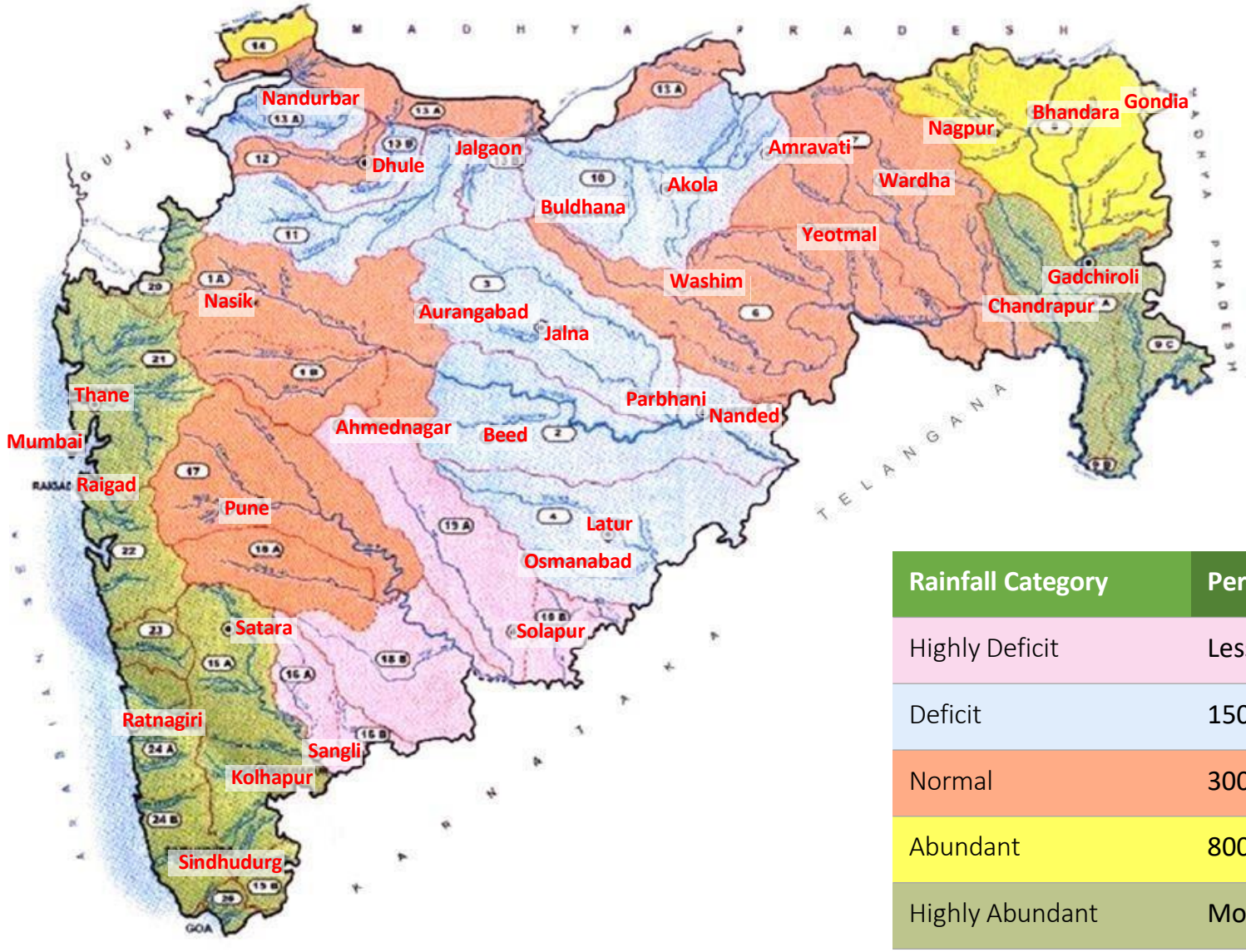
Water Resource Availability in Maharashtra



Rainfall Category	Per ha. Availability	Areal Distribution
Highly Deficit	Less than 1500 m³	10.50%
Deficit	1500 m³ to 3000 m³	32.00%
Normal	3000 m³ to 8000 m³	30.00%
Abundant	8000 m³ to 12000 m³	7.00%
Highly Abundant	More than 12000 m³	20.50%

Vision and key initiatives to achieve it

Water access

All people with secured access to potable >55 and >135 LPCD in rural and urban areas resp.

Better irrigation outcomes

65% gross irrigated area as percentage of total cropped area

Strengthening institutional mechanisms

>100% cost recovery of water supply operations across all districts

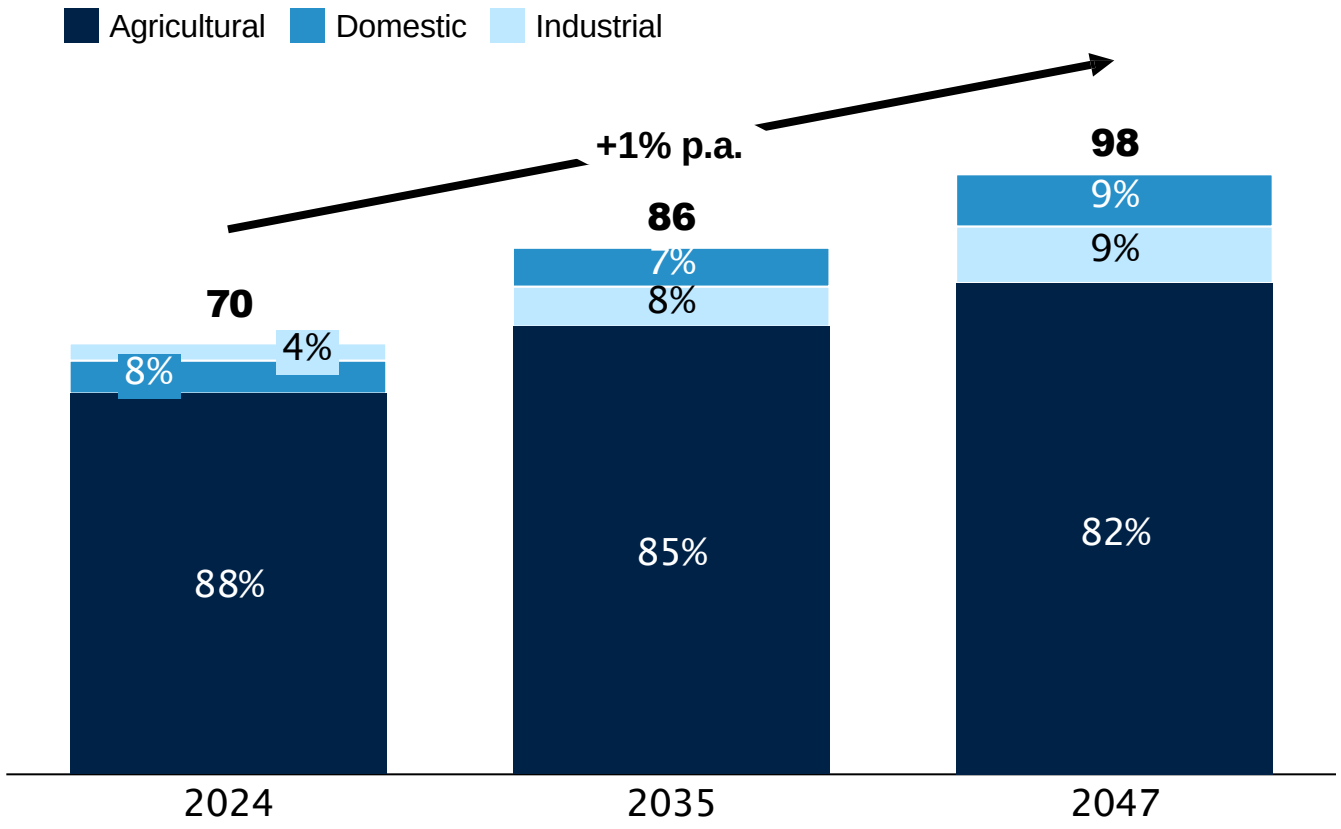
Water circularity

80% wastewater to be reused

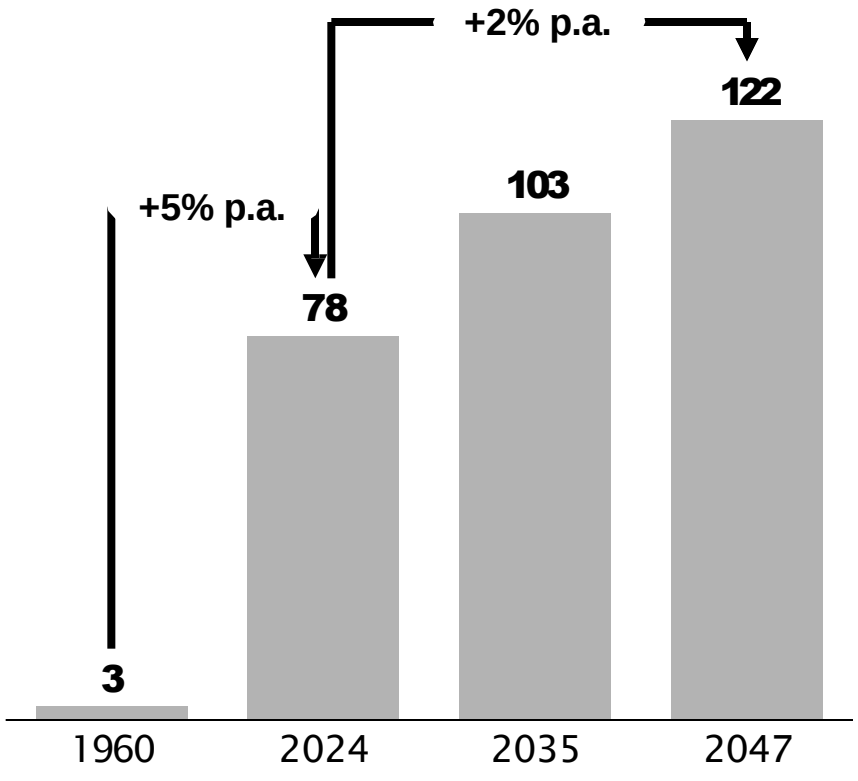
1. **No critical, Semi critical, exploited zone in state**
2. **Expand irrigation coverage enabled by micro-irrigation and modernization**
3. **Tanker free Maharashtra**

Exhibit: Projected demand for water across sectors

Water demand (Billion cubic meters, 2024-47)



Growth in irrigation potential created (excluding ground water, LHa.)



Source: Economic Survey of Maharashtra FY25, FY20; Integrated State Water Plan 2024; NITI Aayog; UN Water

Key initiatives

- Mapping and Ground Truthing of existing structures
- Repair and Rejuvenation of existing structures
- Creating integrated portal for department.
- Develop integrated water security plans

- Minimize demand-supply gaps through recharge of ground and surface water and increase in water recycling and reuse levels

- **Mandate piped distribution network instead of open canal distribution system across the state**

- Modernize and rehabilitate aging irrigation projects, including automating controls, and integrating predictive maintenance
- Improve system performance of distribution network, through on farm development works and efficient irrigation practices
- **Expand coverage of micro-irrigation network: Incentivize multiple private partners to develop micro- irrigation systems at scale. Increase linking with treated wastewater and harvested rainwater**
- **Improve water use efficiency in agriculture through use of soil moisture using sensors, low-capacity pumps in farmers**
- **Digitize water systems with smart meters, GIS, and AI**

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Metrics

Themes	Metric	Current	2029	2035	2047
Water Conservation	Proportion of semi-critical / critical / over exploited GW blocks (%)	16	10	8	0
	Area treatment in Ha.				
	Control of soil erosion by soil conservation				
Better irrigation outcomes	Irrigation potential created (L Ha., including natural and recycled water)	21	23	30	35
	Irrigation potential utilization (L Ha.)				
	Area under micro-irrigation (L Ha.)	23	30	50	78

Source: Jal Jeevan Mission. Central Pollution Control Board, Integrated State Water Plan, Water Resources Department, Water Conservation Department, Directorate of Economics and Statistics

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THANK YOU!

Key details of the initiatives (1/4)

1

Ensure drinking water access from tap for all households

- Develop integrated water security plans across local bodies
- **Ensure stringent water quality standards and monitoring throughout the supply chain, with real-time water quality reports available for users**
- Minimize demand-supply gaps through recharge of ground and surface water and increase in water recycling and reuse levels
- Introduce alternate water supply technologies (e.g., desalination across the Konkan coast)